RAM, GPU, and Storage Troubleshooting

1. RAM Reseating Procedure

To remove and reinstall the RAM module and verify proper detection.

Steps Performed

1. Shut down the PC.
2. Unplugged the power cable and all peripherals.
3. Pressed the power button for 10 seconds to discharge remaining power.
4. Opened the side panel of the computer cabinet.
5. Located the RAM slots on the motherboard.
6. Released the locking clips on both sides of the RAM slot.
7. Carefully removed the RAM module.
8. Inspected the RAM contacts and slot for dust or damage.
9. Reinserted the RAM firmly until both locking clips clicked into place.
10. Closed the cabinet and reconnected the power cable.
11. Powered on the PC.
12. Verified that the BIOS and operating system detected the correct amount of RAM.

Observations

- RAM module seated correctly.
- System booted normally.
- BIOS displayed the installed RAM capacity correctly.
- No POST errors or beep codes were observed.

2. GPU Troubleshooting Checklist

Step	Action	Purpose
1	Identify symptoms (no display, artifacts, crashes, low FPS)	Confirm GPU-related issue
2	Check monitor cable and power connections	Eliminate loose cable problems

Step	Action	Purpose
3	Reseat the graphics card in the PCIe slot	Ensure proper physical connection
4	Update or reinstall GPU drivers	Fix software or driver issues
5	Monitor GPU temperature	Detect overheating
6	Test with another monitor or cable	Rule out display problems
7	Test the GPU in another PC (if available)	Confirm hardware fault
8	Stress-test the GPU	Verify stability under load

Confirming the Fix

- Display works normally.
- No artifacts or screen flickering.
- Drivers install successfully.
- Stress tests complete without crashes.
- Gaming performance returns to normal.

3. Storage Device Reconnection (SATA/NVMe)

Objective

Disconnect and reconnect the storage device to observe BIOS behavior.

Before Reconnecting

The storage drive was disconnected.

BIOS Observation

- SSD/HDD not detected.
- Boot device missing.
- Possible messages:
  - No Boot Device Found
  - Boot Device Not Detected
  - Insert Boot Media

    - Operating System Not Found

After Reconnecting

1. Powered off the system.
2. Reconnected the SATA data and power cables (or reinstalled the NVMe SSD).
3. Restarted the PC.
4. Entered BIOS.

BIOS Observation

- Storage drive detected successfully.
- Windows booted normally.
- No boot errors appeared.

4. Random Restart Troubleshooting

Component Checked First

RAM

Reason

Faulty or improperly seated RAM is one of the most common causes of random restarts, system crashes, and blue screens.

Diagnostic Steps

1. Check that the RAM is seated properly.
2. Clean the RAM contacts if necessary.
3. Test one RAM stick at a time.
4. Run the Windows Memory Diagnostic tool.
5. Run MemTest86 for a thorough memory test.
6. Check BIOS for correct RAM speed and XMP settings.
7. Replace the RAM if errors are detected.

Result

- If no memory errors are found and the system remains stable, the RAM is functioning correctly.

- If errors are detected, replacing the faulty RAM should resolve the random restart issue.